# Advanced Mathematics 2

## Computing and Mathematics

**May, 2026**

setu.ie

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

# Advanced Mathematics 2 (A35245)

| | | | |
|---|---|---|---|
| **Short Title:** | Advanced Mathematics 2 | | |
| **Faculty** | Science and Computing | | |
| **Department** | Computing and Mathematics | | |
| **Cluster** | Mathematics and Physics | | |
| **Author** | ADAVY | | |
| **Credits** | 5 | **Level:** | Introductory |

## Description of Module / Aims

The study of this module enables students to obtain the space analytic geometry, multivariate function calculus and its application, infinite series and ordinary differential equations, and so on the basic concepts, basic theory and basic computing skills, mathematical knowledge for learning the follow-up courses and further expand and improve mathematical literacy and necessary mathematical basis.

## Programmes

| | | |
|---|---|---|
| MATH-0031 | BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI) | 1 / 2 / M |
| MATH-0031 | BEng (Hons) in Information Engineering (International) (WD_EEELC_BI) | 1 / 2 / M |
| MATH-0031 | BSc (Hons) in Applied Computing (International) (WD_KACMI_B) | 1 / 2 / M |
| MATH-0031 | BSc (Hons) in Software Engineering (WD_KDEVP_BI) | 1 / 2 / M |
| MATH-0031 | BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI) | 1 / 2 / M |

## Indicative Content

- Space analytic geometry: plane and its equation, spatial line and its equation, surface and its equation, space curve and its equation, some quadric surfaces.
- Differential calculus of functions of several variables: the concept of multivariate function, partial derivatives, total differential, multivariate function derivative method, the derivation of implicit function formula, multivariate function differential calculus geometry application, directional derivative and gradient, extreme value of a function of many variables.
- Integration of functions of several variables: the concepts, properties, calculation methods and applications of double integral and triple integral
- Curvilinear integral and surface integral: the concepts, properties, calculation methods of two types of curvilinear integral and surface integral, Green formula, Gaussian formula, Stokes' theorem, flux and divergence, loop flow and curl.
- Infinite series: the concept and properties of series of constant series, the convergence method of series of constant series, power series and its expansion, Fourier series.
- Ordinary differential equation: first-order differential equation, high-order linear differential equations, linear differential equation with constant coefficients, the application of differential equations.

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Demonstrate problem-solving skills.
2. Transform the spatial geometry problem into algebra problem through vector and coordinate system, , and improve the spatial imagination ability and operation ability, cultivate the comprehensive problem solving ability through the quadric surface to carry on the subject association.
3. Learn the theory of functions with several variables, including the limit, continuity, differentiation, Taylor expansion, extreme and so on, find and classify extreme of a function of multiple variables, derive the Taylor polynomial of a function of multiple variables.
4. Compute integrals of several variables and integrals for curve and curved surface.
5. Learn the theory of series, including number series and function series.
6. Solve elementary first and second order differential equations.

## Learning and Teaching Methods

- Delivery of the module will be through lectures, tutorials, discussion, and practice.
- The lectures will enable students to obtain the important basic concepts, properties and knowledge system of analytic geometry of space, limit of multivariate function, continuous, total differential, integrals of several variables and integrals for curve and curved surface, infinite series, ordinary differential equation and will develop theory, lead students through worked examples and introduce the context for the module material.
- The tutorial classes will underpin and rehearse the skills covered in lectures. Students will be encouraged, through the problem sheets, to construct valid and precise mathematical arguments and will be expected to produce solutions using appropriate mathematical notation.
- The discussion and practical sessions will be used to discuss applications of the theory and to utilize and implement mathematical software.
- The practical programme is designed to re-enforce the strong interconnections between the mathematical concepts covered in this module and Computer Science concepts using Computer Algebra Systems (CAS) and guided development of students' own code. A typical activity consists of investigating a number of basic quadrature techniques and their application to problems in computer science.

## Assessment Methods

| | Weighing | Outcomes Assessed |
|---|---|---|
| Continuous Assessment | 40% | 1,2,3,4,5 |
| *Assignment* | 20% | 1,2,3,4,5 |
| *In-Class Assessment* | 20% | 1,2,3,4,5 |
| Final Written Examination | 60% | 1,2,3,4,5,6 |

## Assessment Criteria

**<40%:** Inability to understand the concept of functions, complete simple drill exercises in differentiation and integration.

**40%-49%:** Able to demonstrate a basic understanding of the fundamental concepts in calculus as outlined in the indicative content, and apply, to some degree, these concepts to problems in computer science.

**50%-59%:** All the above, in addition to using the standard notation to express mathematical entities.

**60%-69%:** In addition, to be able to determine appropriate mathematical techniques to analyze applied problems and to express their work with rigor and precision.

**70%-100%:** All the before an excellent level. Demonstrates an ability to put a solution into a context and assess whether such solutions are meaningful.

## Learning Modes

| Learning Mode | F/T Hours | P/T Hours |
|---|---|---|
| Lecture | 78 | |
| Tutorial | 12 | |

## Supplementary Material

- "Helping Engineers Learn Mathematics (HELM).." https://www.lboro.ac.uk/departments/mlsc/student-resources/helm-workbooks/
- Banner, A. _Princeton Calculus Reader_. China: People's Posts and Telecommunications Press, 2016.
- Croft, A. and R. Davison. _Foundati__on Maths_. 5th Edition. NY: Pearson, 2010.
- Hughes-Hallet, D. _Calculus_. 6th Edition. NY: Wiley, 2012.
- Shunfeng, W. and et al. _Advanced Mathematics Guidance (Volume 2)_. China: Southeast University Press, 2020.

## Requested Resources

- COMPUTER LAB: Multimedia Lab